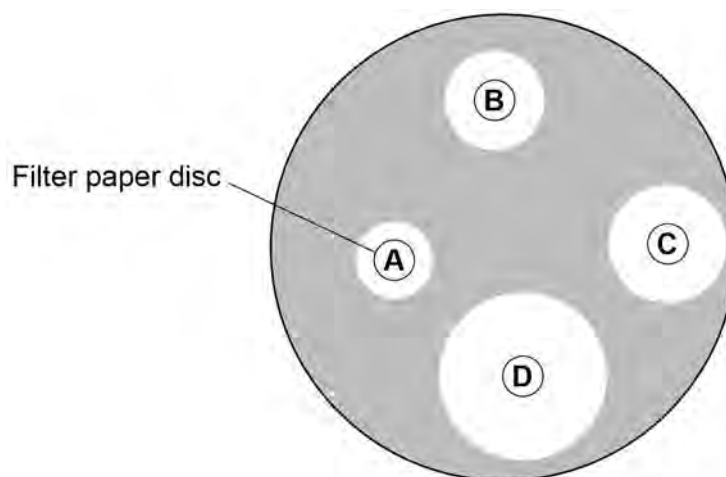


0 8

A student investigated the effectiveness of four different concentrations, **A**, **B**, **C** and **D**, of the same disinfectant on the growth of a bacterium. He grew a culture of this bacterium on nutrient agar (a solid growth medium) in a Petri dish. The student then cut out four filter paper discs and soaked each disc in one of the four concentrations. He then placed the discs on the nutrient agar in the Petri dish. He then left the Petri dish at 25 °C for 24 hours.

Figure 8 shows the appearance of the Petri dish after 24 hours.

Figure 8

**0 8****. 1**

Explain why there is a clear zone around each paper disc.

[1 mark]

0 8**. 2**

The student researched information on this disinfectant prior to carrying out his investigation. On the basis of this research, the student used a maximum concentration of disinfectant of 40%.

Use **Figure 8** to explain why.

[1 mark]



0	8
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3

Suggest **two** variables the student should control in using the filter paper discs in this investigation.

[2 marks]

1 _____

2 _____

0	8
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 .

4

Use the areas of the clear zones in **Figure 8** to determine how many times more effective concentration **D** is than concentration **B**. Show your working.

[2 marks]

Answer = _____

Turn over for the next question

